

sion. The total power consumption of the device is 1000 mW during average use and 60 mW while the device is sleeping.

Logica has applied for multiple patents of the solution. These include technology patents for how the solution works, calculation of the carbon emissions, and the Green Index scores and their end-to-end usage. In addition, business patents for the concept of differential fuel pricing based on emissions have been filed.

Plug-n-play. The EMO has a variety of innovative aspects starting from the device to how the entire solution works seamlessly as a service. The device is extremely easy to use. Simply plug it into the appropriate port of the vehicle and it automatically starts working.

As all the data is wirelessly sent to backend servers, there is no need to manually download any data from the device. The reports are designed keeping in mind the needs of the vehicle drivers and also fleet owners.

"For example, we have developed a unique rating scale called the Logica EMO Green Index score. The Green Index score is an easy-to-understand, quantitative rating scale from 0 to 10 that helps drivers compare their driving with other people's driving as well as chart their progress over time," explains Sanjoy.

Expense control. The device has an inbuilt small computer that does most of the calculations on the device. This helps to keep telecommunication costs (data charges) low by transmitting only the final results to the servers instead of all the data collected. Also, the device is completely self-installable, thus avoiding high installation costs. The device design is also cost-saving



The EMO has a SIM card inside which is used to send all the computed data to Logica backend servers using GPRS, where it is analysed and a report is generated (Image courtesy: <http://business.outlookindia.com>)

Recognitions

The EMO has been extensively recognised for its innovation. Recognitions have come in from:

1. The Economist: Top 10 Game Changing Innovations in Climate Change
2. NASSCOM: Top 6 Innovations Toward Advancement of New Technology
3. Golden Peacock Awards: Innovative Product/Service 2011
4. Institute of Engineering and Technology (UK): Top 5 Innovations in Transportation 2011
5. MIT Technology Review TR35 India 2011

as it avoids the use of expensive unnecessary technology and focuses on the core solution of calculating carbon footprint.

What lies ahead

The EMO is also a business platform that provides incentives to drivers to reduce their vehicle emissions. For example, Logica EMO can be used by governments to provide differential fuel pricing. Differential fuel pricing is a unique concept that helps promote eco-friendly driving by passing the incentive of reduced emissions to drivers through lower fuel prices. Data from the EMO device can be transferred to readers at a petrol pump, which can then be used to calculate the price of

fuel in real time based on the carbon footprint of the vehicle.

Governments, through progressive policy making, can introduce laws to charge a lower fuel tax for vehicles with lower carbon footprint, thus providing a monetary incentive to drivers to improve their driving habits. The same concept can also be extended to other types of taxes such as toll tax and road tax and parking fees.

Sanjoy adds, "We are working on bringing in new features such as remote vehicle tracking, theft tracking and geofencing. Geofencing is a virtual perimeter for a real-world geographic area." ●

The author is a tech correspondent at EFM Bengaluru

THE COMPLETE MAGAZINE ON OPEN SOURCE

LINUX

BETA
ForYou

Your favourite Linux Magazine is
now on the Web, too.

LinuxForU.com

Follow us on Twitter @LinuxForYou